

Environmental Sound Classification Using Temporal and Spectral Audio Features

1. Student Information

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2. Project Overview

Environmental sounds contain useful information about surrounding events and acoustic environments. Automatic recognition of these sounds has applications in environmental monitoring and intelligent acoustic systems [3].

This project aims to develop an environmental sound classification system using handcrafted temporal and spectral audio features. Audio recordings will be pre-processed, divided into short-time frames, and represented using features such as RMS energy, zero-crossing rate, spectral centroid, spectral spread, spectral flatness, and spectral rolloff [1].

These features will be used to train conventional classifiers, including k-nearest neighbours (kNN) and support vector machines (SVM). The project will compare temporal features, spectral features, and their combination to investigate how different feature groups affect classification performance.

3. Background and Motivation

Audio content analysis aims to extract meaningful information from sound signals. Feature-based methods represent raw audio using compact and interpretable descriptors before applying machine-learning classifiers [1].

Temporal and spectral features capture complementary properties of sound. Features such as RMS energy, zero-crossing rate, spectral centroid, spectral spread, spectral flatness, and spectral rolloff have been widely used to characterise waveform and frequency-domain information for audio classification [1].

Public datasets support systematic evaluation of environmental sound classification methods. The ESC dataset provides a benchmark for this task, with ESC-50 containing recordings from 50 environmental sound classes [2]. Related UrbanSound research also highlights the value of realistic labelled environmental audio data [3].

Although recent studies have applied deep convolutional neural networks to environmental sound classification [4], this project focuses on handcrafted audio features with conventional classifiers. This approach is computationally manageable, interpretable,

and suitable for investigating how temporal and spectral features influence classification performance.

4. Proposed Methodology

The project will be implemented mainly in MATLAB. Environmental sound recordings will be obtained from the publicly available ESC-50 dataset [2]. ESC-50 provides a standard benchmark for environmental sound classification and contains multiple categories of common real-world sounds [2]. To keep the project feasible, a balanced subset containing approximately five to eight representative sound classes will initially be selected.

Possible classes may include dog barking, rain, crying baby, clock or bell sounds, engine noise, birds, and other representative environmental sounds available in the dataset.

Audio signals will first be converted to mono where necessary and resampled to a common sampling frequency. Amplitude normalisation will also be applied. Each audio clip will then be divided into overlapping short-time frames. A frame duration of approximately 20–30 ms will initially be investigated, with an appropriate hop size between consecutive frames.

For every frame, temporal and spectral features will be calculated. The main features considered will include:

- RMS energy
- Zero-crossing rate
- Spectral centroid
- Spectral spread
- Spectral flatness
- Spectral rolloff

Short-time spectral descriptors are commonly used to characterise timbral and frequency-domain properties of audio signals for classification [1]. The frequency-domain features in this project will be obtained from the short-time Fourier spectrum after windowing each frame.

Since an audio recording contains many frames, frame-level features will be summarised using statistical measures such as the mean and standard deviation. This will produce one fixed-length feature vector for each audio recording.

The dataset will then be divided into training and testing subsets while maintaining approximately equal class proportions. Feature normalisation will be performed using statistics calculated from the training data.

Two conventional classifiers will initially be investigated: k-nearest neighbours and support vector machines. Their performance will be evaluated under three main feature configurations:

1. Temporal features only;

2. Spectral features only;
3. Combined temporal and spectral features.

This comparison will allow the project to investigate which type of signal representation provides the most useful information for environmental sound classification.

Classification performance will be evaluated using overall accuracy, precision, recall, F1-score, and confusion matrices. Feature distributions and selected spectrograms will also be visualised to help explain why particular sound classes are easier or more difficult to distinguish.

5. Expected Outcomes

The expected outcome is a MATLAB-based environmental sound classification system capable of audio pre-processing, feature extraction, model training, and sound-category prediction.

The project will compare temporal, spectral, and combined feature sets, as well as kNN and SVM classifiers. Performance will be evaluated using accuracy, precision, recall, F1-score, and confusion matrices.

Final deliverables will include MATLAB source code, feature and spectrogram visualisations, classification results, project documentation, and example predictions. All materials will be maintained in a public GitHub repository with a README file and GitHub Pages project site.

6. Timeline

Weeks 1–2: Finalise the project scope, select environmental sound categories, create the GitHub repository, and identify the specific subset of ESC-50 to be used.

Weeks 3–5: Conduct the literature review, obtain and organise the ESC-50 dataset [2], and study temporal and spectral audio features relevant to the selected sound classes.

Weeks 6–7: Implement audio loading, resampling, normalisation, framing, windowing, and basic signal visualisation in MATLAB.

Weeks 8–9: Implement RMS, ZCR, spectral centroid, spectral spread, spectral flatness, and spectral rolloff extraction. Generate fixed-length feature vectors for the selected dataset.

Weeks 10–11: Train kNN and SVM classifiers and conduct experiments using temporal-only, spectral-only, and combined feature sets. Evaluate accuracy, precision, recall, F1-score, and confusion matrices.

Weeks 12–13: Analyse the results, prepare final figures and comparisons, complete the final report and README documentation, and finalise the GitHub Pages site.

References

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